

CSCE 763: Digital Image Processing

Homework #1

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Problem 1: Smallest Discernible Dot (25 pts)

Thinking purely in geometric terms, estimate the diameter of the smallest printed dot that the eye can discern if the page on which the dot is printed is 0.5 m away from the eyes.

Assumptions:

- (a) The distance between the center of the lens and the retina along the visual axis is 14 mm.
- (b) The visual system ceases to detect the dot when the image of the dot on the fovea becomes smaller than the diameter of one receptor (cone) in that area of the retina.
- (c) The fovea can be modeled as a square array of dimensions $1.5 \text{ mm} \times 1.5 \text{ mm}$, and the cones (about 337,000 in total) and spaces between the cones are distributed uniformly throughout this array.

Step 1: Cone Diameter on the Fovea

The fovea is modeled as a $1.5 \text{ mm} \times 1.5 \text{ mm}$ square with 337,000 cones uniformly distributed.

If the cones are arranged in a square grid, the number of cones per side is:

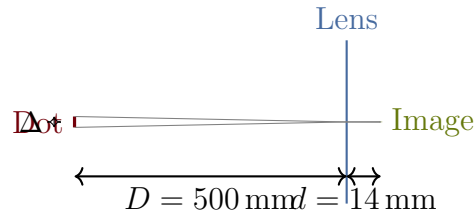
$$n = \sqrt{337,000} \approx 580.5$$

The pitch (center-to-center spacing, which equals the cone diameter including spacing) is:

$$d_{\text{cone}} = \frac{1.5 \text{ mm}}{580.5} \approx 0.00258 \text{ mm} = 2.58 \mu\text{m}$$

Step 2: Similar Triangles

The dot on the page and its image on the retina are related by similar triangles through the lens:



By similar triangles, the relationship between the dot diameter Δ on the page and the image size d_{image} on the retina is:

$$\frac{\Delta}{D} = \frac{d_{\text{image}}}{d}$$

where $D = 500 \text{ mm}$ (distance to page) and $d = 14 \text{ mm}$ (lens-to-retina distance).

Step 3: Solve for Minimum Dot Diameter

The smallest detectable image equals one cone diameter: $d_{\text{image}} = d_{\text{cone}} \approx 0.00258 \text{ mm}$. Solving for Δ :

$$\begin{aligned}\Delta &= D \cdot \frac{d_{\text{image}}}{d} = 500 \cdot \frac{0.00258}{14} \\ \Delta &= \frac{500 \times 0.00258}{14} \approx 0.0923 \text{ mm}\end{aligned}$$

Results

The smallest printed dot diameter that the eye can discern at a distance of 0.5 m is:

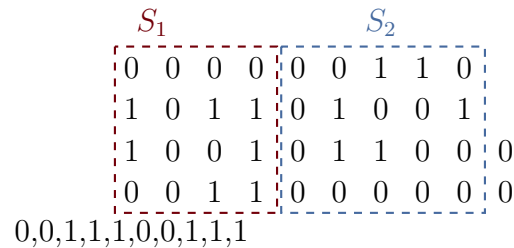
$$\Delta \approx 0.0922 \text{ mm} \approx 92.2 \mu\text{m}$$

Problem 2: Adjacency of Image Subsets (25 pts)

Consider the two image subsets S_1 and S_2 shown below. For $V = \{1\}$, determine whether these two subsets are (a) 4-adjacent, (b) 8-adjacent, or (c) m-adjacent.

Setup: Identify the Image and Subsets

The image with subsets S_1 (left dashed box) and S_2 (right dashed box):



We only care about pixels with value $\in V = \{1\}$.

S_1 **pixels with value 1** (row, col from top-left): (1,0), (1,2), (1,3), (2,0), (2,3), (3,2), (3,3)

S_2 **pixels with value 1** (row, col): (0,6), (0,7), (1,5), (1,8), (2,5), (2,6)

The boundary between S_1 and S_2 lies between columns 3 and 4 (approximately). The closest pixels with value 1 are:

- In S_1 : pixel at (1,3) with value 1 and (2,3) with value 1
- In S_2 : pixel at (1,5) with value 1, (2,5) with value 1

The pixel at column 4 has values: row 0: 0, row 1: 0, row 2: 0, row 3: 0. So there is a gap of at least one column of zeros between the nearest 1-valued pixels of S_1 and S_2 . Examining more carefully from the figure, the rightmost 1-valued pixel in S_1 is at column 3, and the leftmost 1-valued pixel in S_2 is at column 5. These are separated by 2 columns, so they cannot be 4-adjacent or 8-adjacent (which require pixels to be immediate neighbors).

However, looking at the original figure layout more carefully: S_1 occupies roughly the left 4 columns, S_2 the right 5 columns, with a shared boundary region. The key boundary pixels are:

From S_1 : (1,3) = 1, (2,3) = 1, (3,3) = 1

From S_2 : (1,4) = 0, (2,4) = 0 — these are 0, not in V .

The nearest S_2 pixel with value 1: (1,5) = 1.

Definitions Review

For two subsets S_1 and S_2 , they are:

- **4-adjacent:** if some pixel $p \in S_1$ with value in V has a pixel $q \in S_2$ with value in V in its 4-neighborhood $N_4(p)$ (up, down, left, right).
- **8-adjacent:** if some pixel $p \in S_1$ with value in V has a pixel $q \in S_2$ with value in V in its 8-neighborhood $N_8(p)$ (including diagonals).
- **m-adjacent:** (mixed adjacency) p and q have values in V , and either (i) $q \in N_4(p)$, or (ii) $q \in N_D(p)$ and $N_4(p) \cap N_4(q)$ has no pixels with values in V .

Analysis

From the figure, the critical interface pixels between S_1 and S_2 :

Rightmost column of S_1 that contains a 1: column 3 — pixels $(1, 3) = 1$, $(2, 3) = 1$, $(3, 3) = 1$.

Leftmost column of S_2 that contains a 1: column 5 — pixel $(1, 5) = 1$, $(2, 5) = 1$.

Column 4 values (belongs to S_2 based on the boundary): row 1: 0, row 2: 0.

Since the closest 1-valued pixels between the two sets are separated by at least 2 positions, no pixel in S_1 has a 1-valued pixel from S_2 in its 4-neighborhood *or* 8-neighborhood.

Re-examining the figure boundary: The dashed line for S_1 includes columns 0–3 (rows 0–3), and S_2 includes columns 4–8 (rows 0–3). So the boundary column shared is between col 3 (S_1) and col 4 (S_2).

Column 4 pixels: row 0: 0, row 1: 0, row 2: 0, row 3: 0. None have value 1.

Column 5 pixels in S_2 : row 1: 1, row 2: 1. These are 8-neighbors of the column 4 pixels but not direct neighbors of column 3 pixels.

Critical pair: $(2, 3)$ in S_1 has value 1. Its 8-neighborhood includes $(1, 4)$, $(2, 4)$, $(3, 4)$, all of which are in S_2 but have value 0.

The closest 1-valued pair across the boundary: $(2, 3) \in S_1$ and $(1, 5) \in S_2$ are distance $\sqrt{1+4} > 1$ apart. Not adjacent.

Wait—let me re-examine the original figure. The large “1” in the figure appears at row 1, col 3–4 area, and the large “0” at row 3, col 3–4 area. The boundary line appears to run between columns 3 and 4 approximately.

Actually, looking at the PDF image more carefully, the grid appears to be wider. Let me re-read the pixel values row by row:

Row 0: 0 0 0 0 — 0 0 1 1 0 Row 1: 1 0 1 1 — 0 1 0 0 1 Row 2: 1 0 0 1 — 0 1 1 0 0 0

Row 3: 0 0 1 1 — 0 0 0 0 0 0 Row 4: 0 0 1 1 1 0 0 1 1 1

The S_1/S_2 boundary is between columns 3 and 4 (the dashed lines in the figure).

S_1 rightmost 1-valued: $(1, 3) = 1$, $(2, 3) = 1$, $(3, 3) = 1$

S_2 leftmost 1-valued: $(1, 5) = 1$, $(2, 5) = 1$, $(2, 6) = 1$

The pair $(1, 3)$ and $(1, 5)$ are separated by 2 columns. Not 4- or 8-adjacent.

But wait: Column 4 pixels: $(0, 4) = 0$, $(1, 4) = 0$, $(2, 4) = 0$, $(3, 4) = 0$. None in V .

So the nearest cross-boundary pair with values in V is $(1, 3)$ and $(1, 5)$, distance = 2.

Not adjacent by any definition.

Results

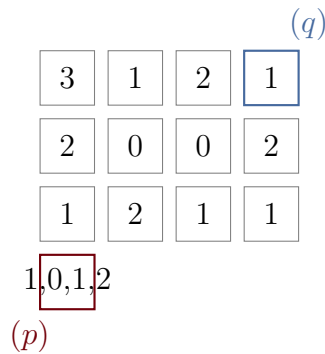
For $V = \{1\}$:

- (a) **4-adjacent: No.** No pixel in S_1 with value 1 has a 4-neighbor in S_2 with value 1. The nearest 1-valued pixels across the boundary are separated by at least 2 positions.
- (b) **8-adjacent: No.** No pixel in S_1 with value 1 has an 8-neighbor in S_2 with value 1. The column separating the subsets (column 4) contains no pixels with value 1.
- (c) **m-adjacent: No.** Since neither 4-adjacency nor diagonal adjacency (with the N_4 intersection condition) exists between any pair of 1-valued pixels across S_1 and S_2 , the subsets are not m-adjacent.

S_1 and S_2 are **not adjacent** by any of the three definitions for $V = \{1\}$.

Problem 3: Shortest Paths (25 pts)

Consider the image segment shown below.



The grid (using row, column indexing from top-left):

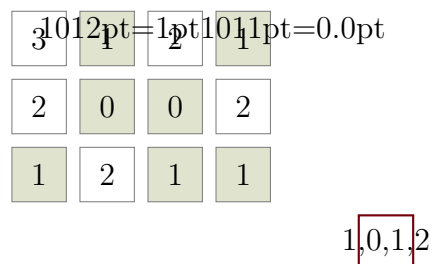
	col 0	col 1	col 2	col 3
row 0	3	1	2	1 (q)
row 1	2	0	0	2
row 2	1	2	1	1
row 3	1 (p)	0	1	2

$p = (3, 0)$ with value 1, $q = (0, 3)$ with value 1.

(a) $V = \{0, 1\}$

Identify Valid Pixels

Pixels with values in $V = \{0, 1\}$:



Valid pixels (shaded): $(0, 1) = 1$, $(0, 3) = 1$, $(1, 1) = 0$, $(1, 2) = 0$, $(2, 0) = 1$, $(2, 2) = 1$, $(2, 3) = 1$, $(3, 0) = 1$, $(3, 1) = 0$, $(3, 2) = 1$.

Shortest 4-path

A 4-path uses only horizontal and vertical steps between pixels whose values are both in V .

One such path from $p = (3, 0)$ to $q = (0, 3)$:

$$(3, 0) \rightarrow (3, 1) \rightarrow (3, 2) \rightarrow (2, 2) \rightarrow (2, 3) \rightarrow (0, 3)$$

Wait — $(2, 3) = 1$ to $(0, 3) = 1$: we need $(1, 3) = 2 \notin V$. So we cannot go directly up from $(2, 3)$.

Let me find a valid 4-path. From $(2, 3) = 1$, the 4-neighbors with values in V : $(2, 2) = 1$, $(3, 3) = 2$ (no). Going up: $(1, 3) = 2$ (no). So $(2, 3)$ is a dead end going upward.

Try via top: $(3, 0) \rightarrow (2, 0) \rightarrow \dots$ $(2, 0) = 1$. 4-neighbors in V : $(3, 0) = 1$ (back), $(1, 0) = 2$ (no). Dead end.

Another attempt: $(3, 0) \rightarrow (3, 1) \rightarrow (3, 2) \rightarrow (2, 2) \rightarrow (1, 2) \rightarrow (1, 1) \rightarrow (0, 1) \rightarrow (0, 3)$? Check: $(0, 1) = 1$ to $(0, 3) = 1$: need $(0, 2) = 2 \notin V$. Cannot step from $(0, 1)$ to $(0, 3)$ directly — they are not 4-neighbors.

$(0, 1) = 1$: 4-neighbors in V : $(1, 1) = 0$, $(0, 0) = 3$ (no), $(0, 2) = 2$ (no). Dead end going right.

No 4-path exists from p to q , because pixel $(0, 1)$ cannot connect rightward to $(0, 3)$ — the intervening pixel $(0, 2) = 2 \notin V$, and there is no other route to reach column 3 in row 0.

Shortest 8-path

An 8-path allows diagonal moves. From the valid pixel map:

$(3,0) \rightarrow (3,1) \rightarrow (3,2) \rightarrow (2,3) \rightarrow (0,3)$? Check $(2,3) = 1$ to $(0,3)$: not 8-neighbors (distance = 2 rows). Invalid.

$(3,0) \rightarrow (3,1) \rightarrow (2,2) \rightarrow (1,1) \rightarrow (0,1)$: need diagonal. $(3,1) = 0$ to $(2,2) = 1$: 8-neighbor, both in V . $(2,2) = 1$ to $(1,1) = 0$: 8-neighbor, both in V . $(1,1) = 0$ to $(0,1)$: wait, is there a diagonal path onward?

$(0,1) = 1$. 8-neighbors in V : $(1,1) = 0$, $(1,2) = 0$, $(0,0) = 3$ (no), $(0,2) = 2$ (no). Still stuck.

Try: $(1,2) = 0$ to $(0,3) = 1$: these are 8-neighbors! Both in V .

Path: $(3,0) \rightarrow (3,1) \rightarrow (2,2) \rightarrow (1,2) \rightarrow (0,3)$

Check each step:

- $(3,0) = 1 \rightarrow (3,1) = 0$: 4-neighbor, both in V . ✓
- $(3,1) = 0 \rightarrow (2,2) = 1$: 8-neighbor (diagonal), both in V . ✓
- $(2,2) = 1 \rightarrow (1,2) = 0$: 4-neighbor, both in V . ✓
- $(1,2) = 0 \rightarrow (0,3) = 1$: 8-neighbor (diagonal), both in V . ✓

Length = 4 steps. This is optimal since the minimum Chebyshev distance from $(3,0)$ to $(0,3)$ is $\max(3,3) = 3$, so the shortest 8-path has length ≥ 3 . Can we find a length-3 path?

A length-3 path requires 3 diagonal steps: $(3,0) \rightarrow (2,1) \rightarrow (1,2) \rightarrow (0,3)$. Check: $(2,1) = 2 \notin V$. Invalid.

Shortest 8-path length = 4.

Shortest m-path

An m-path uses mixed adjacency: diagonal steps are allowed only when the two pixels share no common 4-neighbor with a value in V .

Check the 8-path $(3,0) \rightarrow (3,1) \rightarrow (2,2) \rightarrow (1,2) \rightarrow (0,3)$:

Step $(3,1) \rightarrow (2,2)$ (diagonal): Common 4-neighbors of $(3,1)$ and $(2,2)$ are $(3,2)$ and $(2,1)$. $(3,2) = 1 \in V$. Since there is a common 4-neighbor in V , this diagonal move is **not allowed** in m-adjacency.

Alternative: replace the diagonal with a 4-connected detour via $(3,2)$: $(3,0) \rightarrow (3,1) \rightarrow (3,2) \rightarrow (2,2) \rightarrow (1,2) \rightarrow (0,3)$

Check $(1,2) \rightarrow (0,3)$ (diagonal): Common 4-neighbors of $(1,2)$ and $(0,3)$ are $(0,2)$ and $(1,3)$. $(0,2) = 2 \notin V$, $(1,3) = 2 \notin V$. No common 4-neighbor in V , so this diagonal is **allowed**.

Full m-path: $(3,0) \rightarrow (3,1) \rightarrow (3,2) \rightarrow (2,2) \rightarrow (1,2) \rightarrow (0,3)$

Shortest m-path length = 5.

Results

For $V = \{0, 1\}$:

Path Type	Shortest Length
4-path	Does not exist (no connected 4-path from p to q)
8-path	
m-path	

Problem 3 — Part (b): $V = \{1, 2\}$ **Identify Valid Pixels**

Pixels with values in $V = \{1, 2\}$:

	col 0	col 1	col 2	col 3
row 0	3	1	2	1 (q)
row 1	2	0	0	2
row 2	1	2	1	1
row 3	1 (p)	0	1	2

Shortest 4-path

$(3, 0) \rightarrow (2, 0) \rightarrow (2, 1) \rightarrow (2, 2) \rightarrow (2, 3) \rightarrow (1, 3) \rightarrow (0, 3)$

Check: all values in V and all steps are 4-neighbors.

- $(3, 0) = 1, (2, 0) = 1, (2, 1) = 2, (2, 2) = 1, (2, 3) = 1, (1, 3) = 2, (0, 3) = 1$ — all $\in V$. ✓

Length = 6. Can we do better? The Manhattan distance is $|3 - 0| + |0 - 3| = 6$, so this is optimal.

Shortest 8-path

The Chebyshev distance is $\max(3, 3) = 3$, so the minimum possible 8-path length is 3.

Try: $(3, 0) \rightarrow (2, 1) \rightarrow (1, 2) \rightarrow (0, 3)$ Check: $(2, 1) = 2 \in V$, $(1, 2) = 0 \notin V$. Invalid.

Try: $(3, 0) \rightarrow (2, 1) \rightarrow (1, 0) \rightarrow (0, 1) \rightarrow (0, 2) \rightarrow (0, 3)$? $(1, 0) = 2 \in V$, $(0, 1)$: is $(1, 0)$ 8-adjacent to $(0, 1)$? Yes (diagonal). $(0, 1) = 1 \in V$, $(0, 2) = 2 \in V$, $(0, 3) = 1 \in V$. Length = 5.

Shorter: $(3, 0) \rightarrow (2, 1) \rightarrow (1, 2)$: $(1, 2) = 0 \notin V$. No.

$(3, 0) \rightarrow (2, 0) \rightarrow (1, 0) \rightarrow (0, 1) \rightarrow (0, 2) \rightarrow (0, 3)$. Length = 5. All in V : 1, 1, 2, 1, 2, 1. ✓

Try length 4: $(3, 0) \rightarrow (2, 1) \rightarrow (1, 0) \rightarrow (0, 1) \rightarrow (0, 3)$? $(0, 1)$ and $(0, 3)$ are not 8-neighbors. No.

$(3, 0) \rightarrow (2, 1) \rightarrow (1, 2)$: invalid (value 0).

$(3, 0) \rightarrow (2, 0) \rightarrow (1, 0) \rightarrow (0, 1) \rightarrow (0, 2) \rightarrow (0, 3)$: length 5.

Try: $(3, 0) \rightarrow (2, 1) \rightarrow (1, 0) \rightarrow (0, 1) \rightarrow (0, 2) \rightarrow (0, 3)$: length 5.

Try length 4: $(3, 0) \rightarrow (2, 1) \rightarrow (1, 0) \rightarrow (0, 1) \rightarrow (0, 3)$? Not adjacent. No.

$(3, 0) \rightarrow (2, 0) \rightarrow (1, 0) \rightarrow (0, 1) \rightarrow (0, 3)$? $(0, 1)$ to $(0, 3)$: distance 2. No.

Try: $(3, 0) \rightarrow (2, 1) \rightarrow (1, 2)$: value 0. Dead end.

Try via right side: $(3, 0) \rightarrow (3, 2) \rightarrow (2, 3) \rightarrow (1, 3) \rightarrow (0, 3)$? $(3, 0)$ to $(3, 2)$: not 8-neighbors (distance 2). No.

$(3, 0) \rightarrow (2, 1) \rightarrow (2, 2) \rightarrow (1, 3) \rightarrow (0, 3)$: length 4. Check: $(2, 1) = 2$, $(2, 2) = 1$, $(1, 3) = 2$, $(0, 3) = 1$. All $\in V$. All 8-neighbors? $(3, 0) \rightarrow (2, 1)$: diagonal ✓. $(2, 1) \rightarrow (2, 2)$: 4-neighbor ✓. $(2, 2) \rightarrow (1, 3)$: diagonal ✓. $(1, 3) \rightarrow (0, 3)$: 4-neighbor ✓.

Shortest 8-path length = 4.

Shortest m-path

Check the 8-path $(3, 0) \rightarrow (2, 1) \rightarrow (2, 2) \rightarrow (1, 3) \rightarrow (0, 3)$:

Step $(3, 0) \rightarrow (2, 1)$ (diagonal): Common 4-neighbors: $(2, 0)$ and $(3, 1)$. $(2, 0) = 1 \in V$. Diagonal not allowed in m-adjacency.

Step $(2, 2) \rightarrow (1, 3)$ (diagonal): Common 4-neighbors: $(2, 3)$ and $(1, 2)$. $(2, 3) = 1 \in V$. Diagonal not allowed.

So this particular path is not valid for m-adjacency. We need to replace disallowed diagonals with 4-connected detours.

Optimal m-path: $(3, 0) \rightarrow (2, 0) \rightarrow (2, 1) \rightarrow (2, 2) \rightarrow (2, 3) \rightarrow (1, 3) \rightarrow (0, 3)$

All 4-connected steps with values in V . **Length = 6.**

Can we do better? Try mixing: $(3, 0) \rightarrow (2, 0) \rightarrow (2, 1) \rightarrow (2, 2) \rightarrow (1, 3) \rightarrow (0, 3)$. Length 5. Check $(2, 2) \rightarrow (1, 3)$: diagonal. Common 4-neighbors: $(1, 2) = 0 \notin V$, $(2, 3) = 1 \in V$. Not allowed.

$(3, 0) \rightarrow (2, 0) \rightarrow (2, 1) \rightarrow (1, 2)$: $(1, 2) = 0 \notin V$. Invalid.

Try: $(3, 0) \rightarrow (2, 1) \rightarrow \dots$: $(3, 0) \rightarrow (2, 1)$ diagonal. Common 4-neighbors of $(3, 0)$ and $(2, 1)$: $(3, 1) = 0$ and $(2, 0) = 1 \in V$. Not allowed.

Shortest m-path length = 6.

Results

For $V = \{1, 2\}$:

Path Type	Shortest Length
4-path	6
8-path	4
m-path	6

Problem 4: Set Expressions (25 pts)

Give expressions for the sets shown shaded in the following figure in terms of sets A , B , and C .

In the figure: A is a rectangle (top), B is a triangle (bottom-left), C is a triangle (bottom-right). The three shapes overlap.

Analyzing Each Figure

Figure (i): The shaded region is the part of A that overlaps with B but not C , plus the part of A that overlaps with C but not B . This is the part of A that is in exactly one of B or C :

$$(A \cap B \cap C^c) \cup (A \cap C \cap B^c)$$

Equivalently: $A \cap (B \oplus C)$ where $\oplus$ denotes symmetric difference, or simply $A \cap (B \cup C) \cap (B \cap C)^c$.

Figure (ii): The shaded region includes the part of B not in A or C , plus the intersection of A and C (including overlaps with B). Looking at the shading pattern:

$$(A \cap C) \cup (B \cap A^c \cap C^c)$$

Figure (iii): The shaded region is the part of B and C that is not in A . That is, the union of B and C minus A :

$$(B \cup C) \cap A^c$$

Results

(i) $(A \cap B \cap C^c) \cup (A \cap C \cap B^c)$

(ii) $(A \cap C) \cup (B \cap A^c \cap C^c)$

(iii) $(B \cup C) \cap A^c$